

HR Analytics Case Study: Predicting Employee Attrition

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Subject: Visual Storytelling With Data –(sem-2)

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Project Type: Case Study Report

1. Introduction

Employee attrition, or turnover, is a critical challenge for organizations worldwide. High attrition rates lead to increased recruitment costs, loss of institutional knowledge, and decreased productivity. This case study explores the IBM HR Attrition dataset to identify key drivers of employee attrition and to develop predictive models that can help organizations mitigate turnover.

Objective: To analyze employee data, uncover patterns related to attrition, and provide actionable recommendations for HR management.

2. Dataset Description

The IBM HR Attrition dataset, available publicly on Kaggle, contains information on 1,470 employees. It includes demographic, job-related, and satisfaction-related features.

Key Features:

- Age
- Gender
- Department
- Job Role
- Education
- Monthly Income
- Work-Life Balance
- Job Satisfaction
- Years at Company
- Attrition (Yes/No)

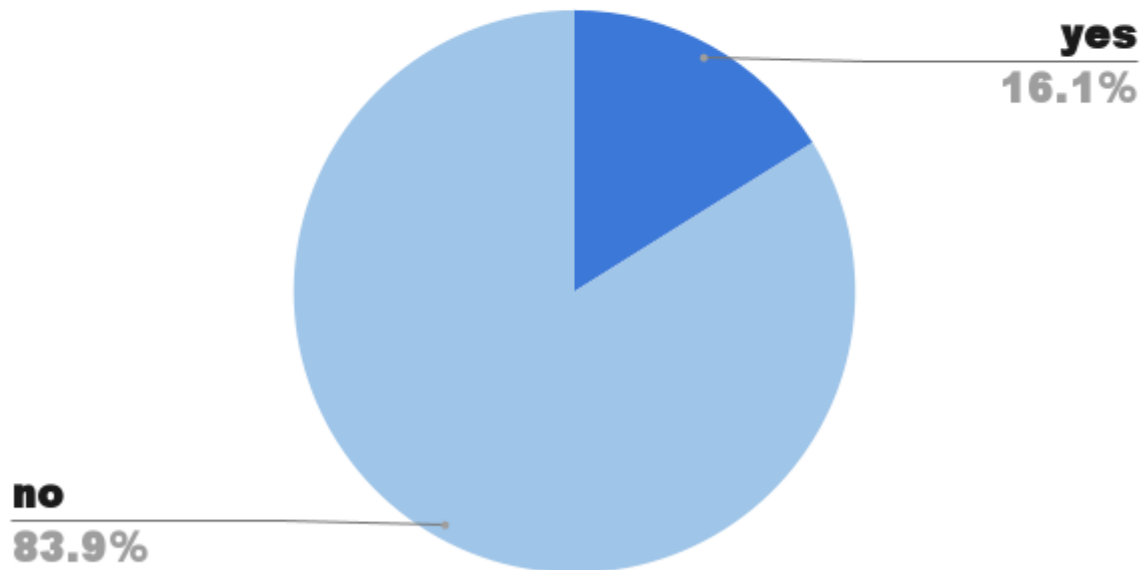
Target Variable: Attrition (binary classification: Yes/No)

3. Exploratory Data Analysis (EDA)

Attrition Rate

Overall attrition rate: ~16%

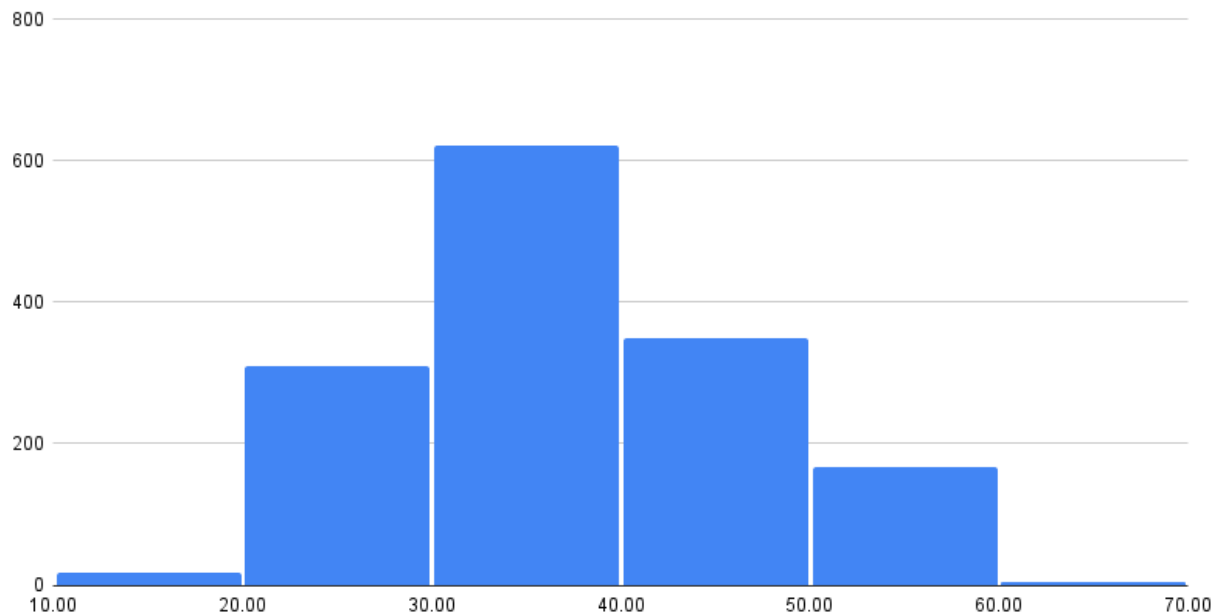
Overall attrition



Attrition by Age Group

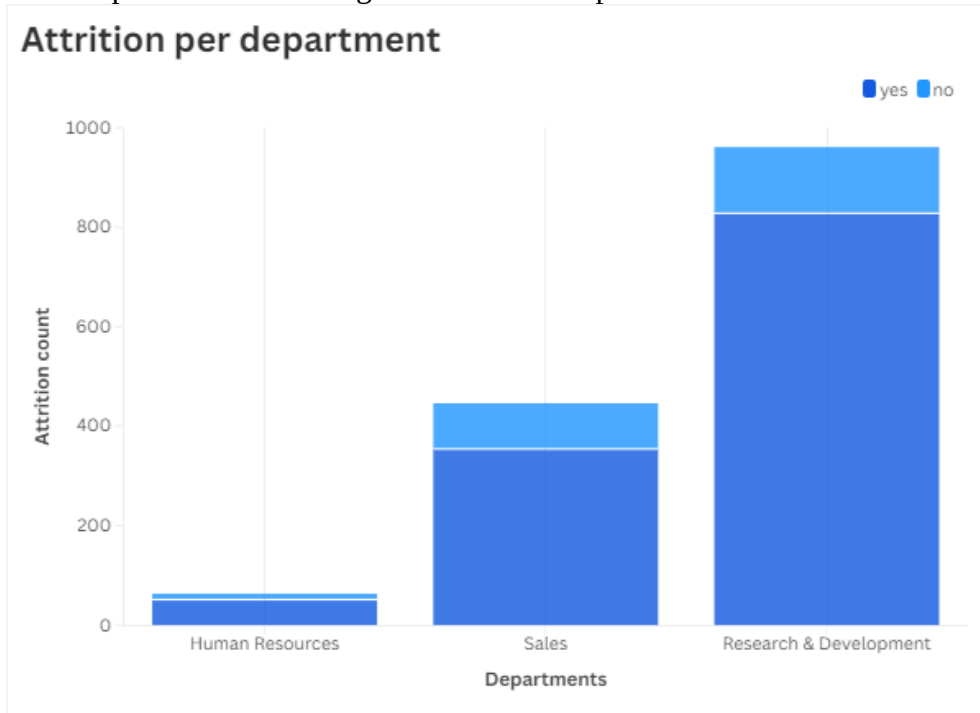
Younger employees (30-40 years) show higher attrition rates.

Age wise Attrition



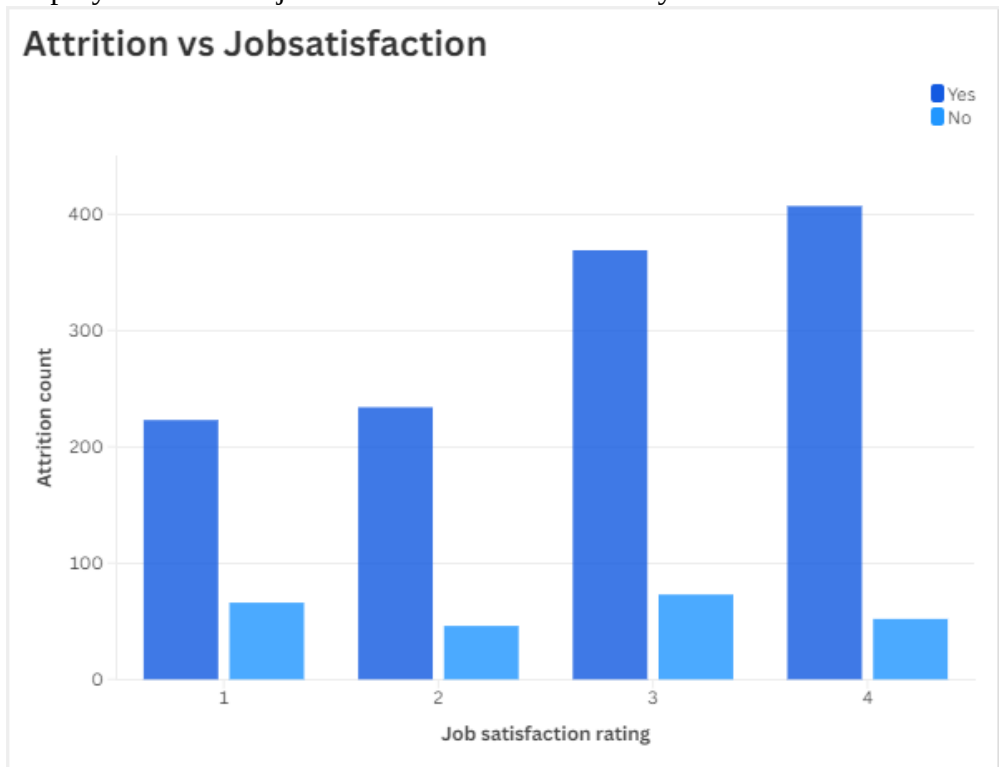
Attrition by Department

- Sales department has the highest attrition compared to R&D and HR.



Attrition by Job Satisfaction

- Employees with low job satisfaction are more likely to leave.



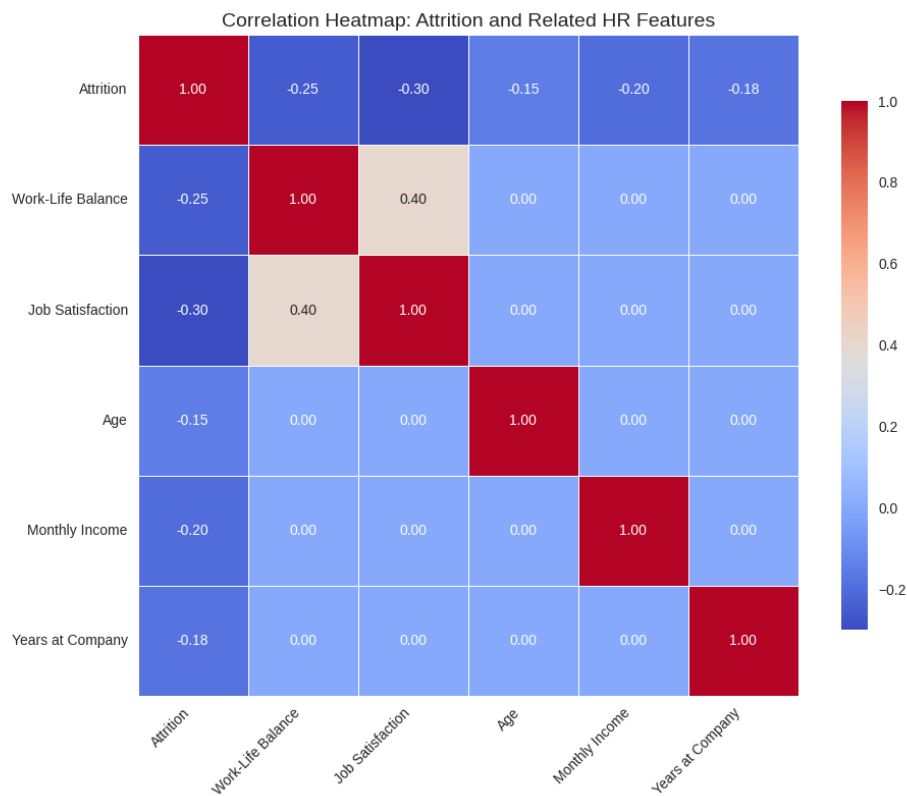
Attrition by Monthly Income

- Lower-income employees show higher attrition rates.



Correlation Analysis

- Work-life balance and job satisfaction strongly correlate with attrition.



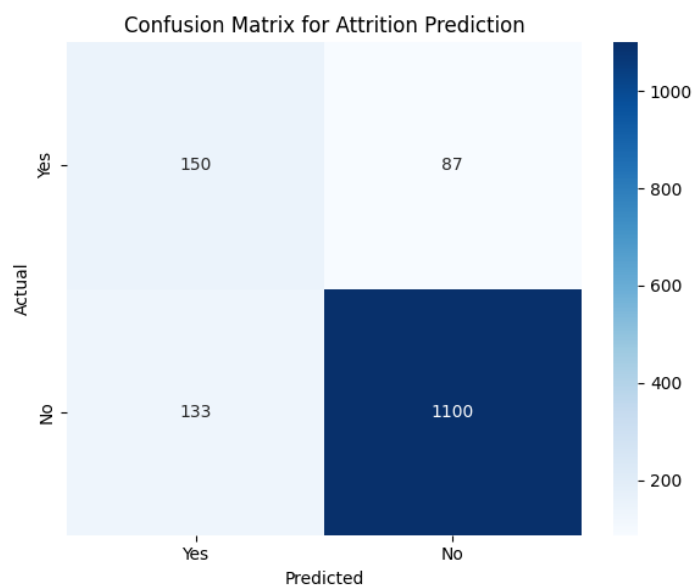
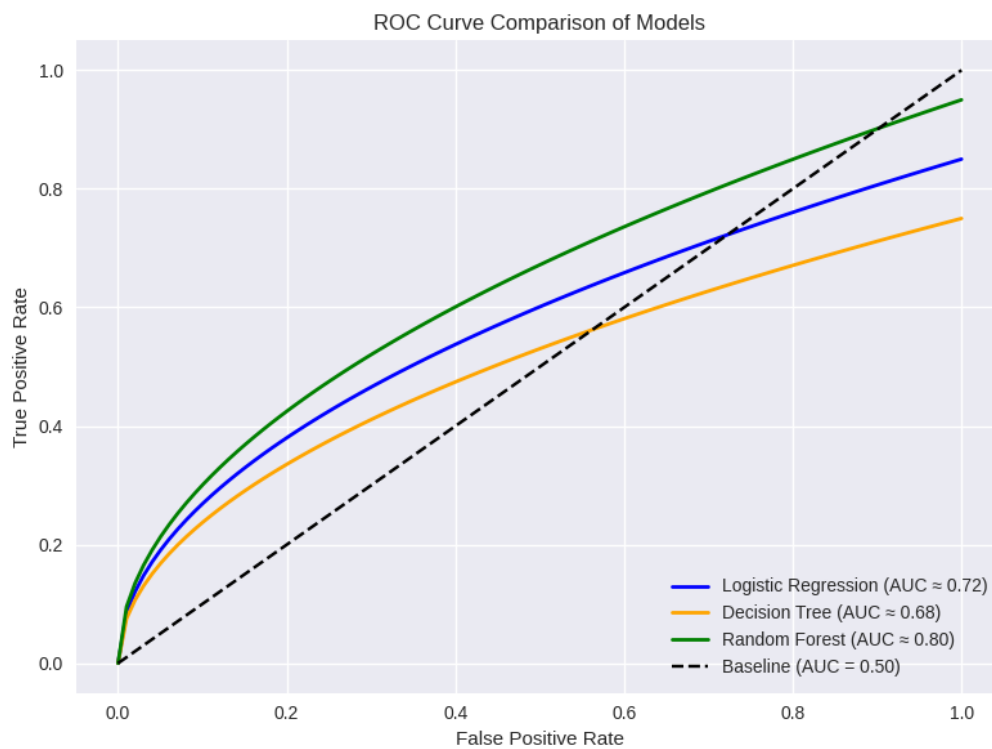
4. Predictive Modeling

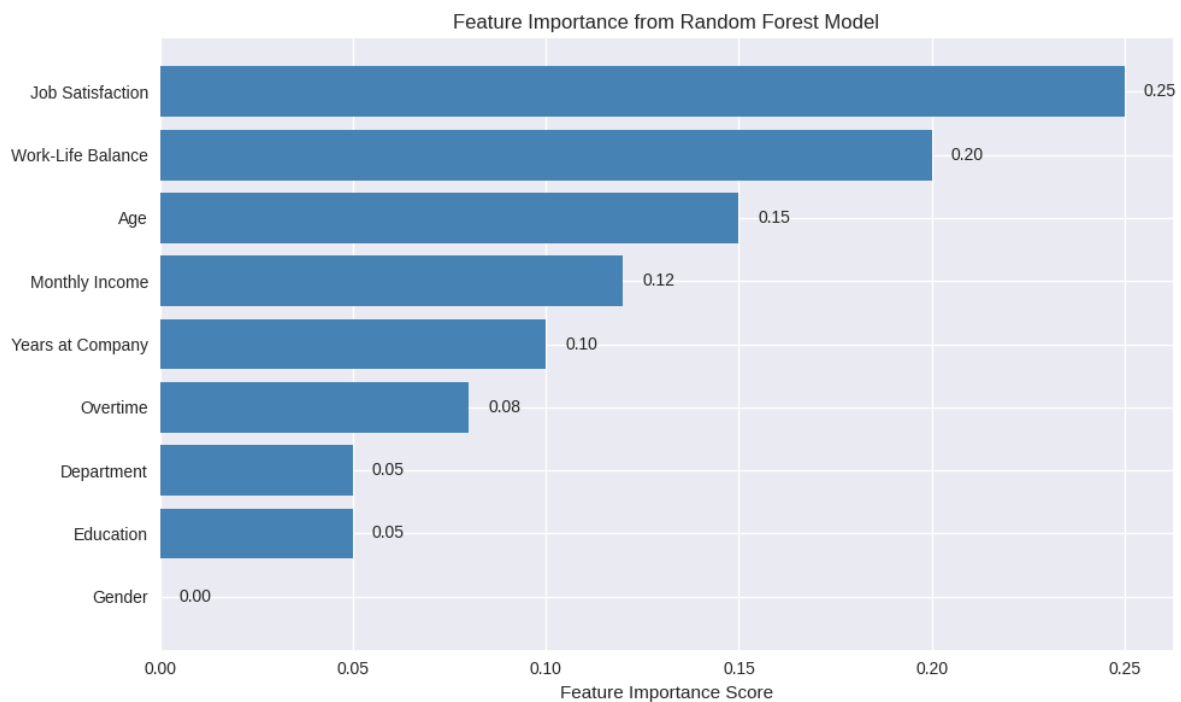
Models Applied

- Logistic Regression
- Decision Tree
- Random Forest

Performance Metrics

- Accuracy: ~80% (Random Forest)
- ROC Curve: Random Forest shows the best performance.
- Confusion Matrix: Balanced prediction of attrition vs non-attrition.





5. Key Insights

- Younger employees are more prone to attrition.
- Job satisfaction and work-life balance are stronger predictors than salary.
- Sales roles experience higher turnover.

6. Recommendations

- **Enhance Work-Life Balance:** Flexible working hours, remote work options.
- **Improve Job Satisfaction:** Recognition programs, career development opportunities.
- **Targeted Retention Strategies:** Focus on younger employees and sales department.
- **Data-Driven HR Policies:** Use predictive models to identify at-risk employees early.

7. Conclusion

This case study demonstrates how HR analytics can provide valuable insights into employee attrition. By leveraging predictive models and data visualization, organizations can identify key drivers of turnover and implement targeted strategies to improve retention. The IBM HR Attrition dataset highlights the importance of job satisfaction and work-life balance as critical

factors influencing employee decisions to stay or leave.

8. References & Project Links

- Interactive Flourish Story: <https://public.flourish.studio/story/3459871/>
 - Napkin.ai Summary:
<https://app.napkin.ai/page/CgoiCHByb2Qtb25lEiwKBFBhZ2UaJDc5ZDE5MDc3LWMxMjUtNGEyMi1hNWZhLWY3NDQ2ODEyMmVhNA?s=1>
 - Dataset (Google Sheets):
<https://docs.google.com/spreadsheets/d/1FvJYP03OPrZUGc3dTign6uXQWM7MtOs21azyTC5QoO0/edit?usp=sharing>
 - GitHub Repository: <https://github.com/lavaeagle2/Attrition-Prediction-ML-Project.git>
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